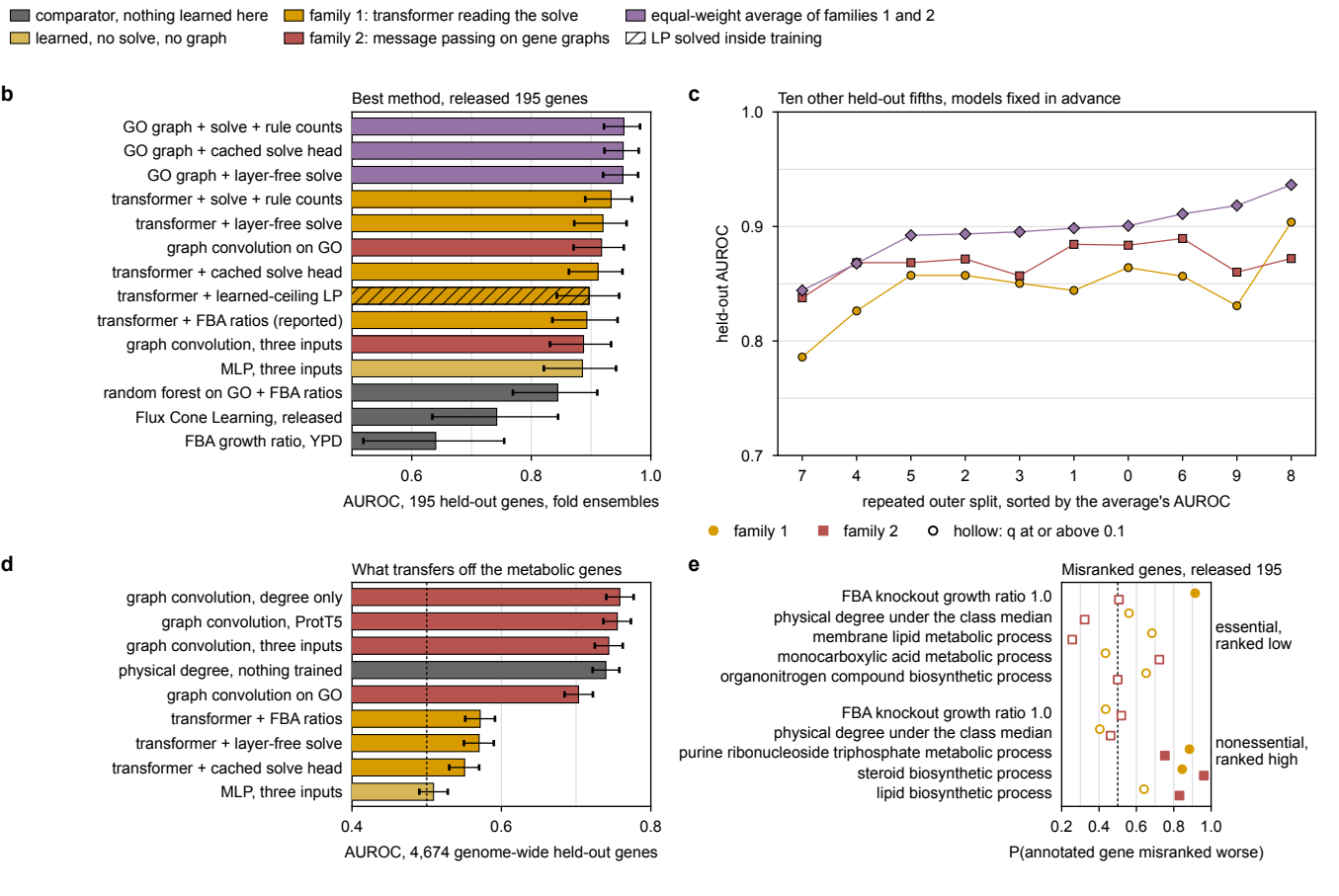
comparator, nothing learned here
learned, no solve, no graph
family 1: transformer reading the solve
family 2: message passing on gene graphs
equal-weight average of families 1 and 2
LP solved inside training
b
Best method, released 195 genes
GO graph + solve + rule counts
GO graph + cached solve head
GO graph + layer-free solve
transformer + solve + rule counts
transformer + layer-free solve
graph convolution on GO
transformer + cached solve head
transformer + learned-ceiling LP
transformer + FBA ratios (reported)
graph convolution, three inputs
MLP, three inputs
random forest on GO + FBA ratios
Flux Cone Learning, released
FBA growth ratio, YPD
0.6
0.8
1.0
AUROC, 195 held-out genes, fold ensembles
c
Ten other held-out fifths, models fixed in advance
held-out AUROC
1.0
0.9
0.8
0.7
7
4
5
2
3
1
0
6
9
8
repeated outer split, sorted by the average's AUROC
family 1
family 2
hollow: q at or above 0.1
d
What transfers off the metabolic genes
graph convolution, degree only
graph convolution, ProtT5
graph convolution, three inputs
physical degree, nothing trained
graph convolution on GO
transformer + FBA ratios
transformer + layer-free solve
transformer + cached solve head
MLP, three inputs
0.4
0.6
0.8
AUROC, 4,674 genome-wide held-out genes
e
Misranked genes, released 195
FBA knockout growth ratio 1.0
physical degree under the class median
membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
FBA knockout growth ratio 1.0
physical degree under the class median
purine ribonucleoside triphosphate metabolic process
steroid biosynthetic process
lipid biosynthetic process
essential, ranked low
nonessential, ranked high
0.2
0.4
0.6
0.8
1.0
P(annotated gene misranked worse)